Consolidated enterprise schema reference v3

71 table definitions and RLS enabled on every table. 58 tables have explicit authenticated SELECT policy/grant; remaining tables are default-deny for ordinary clients. All ordinary client inserts/updates/deletes are denied. This is a complete access script for this reference scope, not a complete implemented enterprise product.

Apply 001_schema.sql, 002_access.sql, 003_permission_seed.sql in a fresh isolated Supabase database as trusted owner. Requires auth.users/auth.uid and anon/authenticated roles. Do not run over existing sx schemas or treat earlier migrations as interchangeable. API exposure is separate; do not expose sx_private. Functions use owner RLS bypass; FORCE RLS is NOT enabled. Superuser/service-role paths remain privileged and require independent checks. Explicit execute grants are included, and all private tables are ungranted.

No migration execution, SQL parser/database test, TypeScript/browser test or production audit was performed. Files generated and checked nonempty only. ERD includes tenant ownership, declared parent links and selected composite relationships; SQL is authoritative for every constraint. Diagram nodes show key fields, not all columns; complete columns are in SQL. Mermaid renderer required. Table inventory and screen matrix are CSV deliverables.

Role definitions/assignments are not automatically seeded. Scope-sensitive permissions use section_id. section.read does not give roster.read. Academic enrollment privacy/official retention must be validated per institution. Personal tables use owner_id and no fake institution tenant; sharing/exports/account suspension are future policies. Raw file references never constitute permission to fetch storage.

Restricted documents require author or explicit audience, whereas business documents can use document.read. Document audience grants expose all document versions in this reference; narrower version-level audiences need separate rules. Contract reference can point to any tenant-owned document version; commands must validate correct document type and executed version. Assigned review/evidence functionality is data storage, not a signing provider or full workflow engine. External signer_reference requires verified party mapping and provider identity checking.

Billing contact ACL is scoped to covered tenant membership; reference read grants are broad for the commercial account, so exposed DTOs/policies must further restrict internal metadata as necessary. Catalog versions are restricted to catalog.read; public pricing needs sanitized endpoints. Financial mutations, state edges, double-entry balance, refund/credit/allocation ceilings, invoice-line reconciliation, merchant ownership and posted immutability need trusted commands. No raw card/CVV fields. Full tax, revenue recognition, aid/tuition, payroll, admissions, degree evaluation, marketplace, housing, localization, AI, retention and institution statutory reporting are not exhaustively modeled.

Preflight: inventory actual deployment, restored backup test, staging fixture identities, constraint/policy checks. Test cross-tenant and cross-section FK rejection; published

parent/content; own submissions/published grades; restricted document audiences; account contact grant expiry; ledger/HR isolation; anonymous/client write denial; storage authorization; helpers owner/access; workflow concurrency/idempotency. Before financial use implement and test command invariants from billing_state_machines.md. Roll forward after posted data; don't drop real financial history.

ERD/screens/specs define reference routes and capability requirements, not deployed handlers. Many commands in the UI matrix remain implementation contracts, not SQL functions in these migrations.